



Sections :A,B,C,D,E,F,G

NMAM INSTITUTE OF TECHNOLOGY, NITTE**Off-Campus Centre of Nitte(Deemed to be University)****II Sem B.Tech. (CBCS) Mid Semester Examinations - I, February 2023****MA1004 -1 – DISCRETE MATHEMATICS AND NUMERICAL METHODS**

Duration: 1 Hour

Max. Marks: 20

Note: Answer any One full question from each Unit.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Encode the message "GERMANY SEEKS AN ALLIANCE" using the key word "NORSE" by using keyword columnar transposition.	2	L*1	1	2
b)	Check whether the following relations on A are reflexive, symmetric or transitive. Justify your answer. (i) $A = \{1, 2, 4, 6, 8\}$, R is a relation on A , defined by aRb if and only if $a \neq b$. (ii) $A = \{a, b, c, d, e\}$ and $R = \{(a, b), (a, c), (a, d), (e, b), (e, d), (e, c)\}$.	4	L3	1	1
c)	Let $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$. Let $f = (3, 1, 6)$ and $g = (4, 8, 2, 5)$. Compute each of the following and write the result as the product of disjoint cycles. (i) $f \circ g$ (ii) $g \circ f$ (iii) $f^{-1} \circ g^{-1}$.	4	L2	1	1
2. a)	Let $U = \{b, d, e, g, h, k, m, n, p, r\}$, $B = \{b, k, m, p, r\}$, $C = \{d, g, m, n, p\}$ and $D = \{d, k, n\}$. Represent the following using bit strings (Use only bit string operations). (i) $B \cup \bar{C}$ (ii) $C \cap D \cap \bar{E}$.	2	L1	1	2
b)	Let $A = \{1, 2, 3, 4\}$. Let R and S be the equivalence relations on A whose matrices are given below. Use Warshall's algorithm to compute the matrix of smallest equivalence relation containing R and S . $M_R = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$ and $M_S = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$.	4	L3	1	2
c)	Let $A = \{1, 2, 3, 4, 5\}$ and R be the relation on A , defined by aRb if and only if $a \leq b + 1$. (i) Find the relation R . (ii) Find the matrix of R and draw the digraph of R . (iii) Find in degree and out degree of each element of A .	4	L2	1	1
Unit – II					
3. a)	State handshaking theorem. Prove that undirected graph has an even number of vertices of odd degree.	5	L1	2	1
b)	Write adjacency matrix and incidence matrix of K_5 and C_4 .	5	L1	2	2
4. a)	Write the following argument in symbolic form and then apply the rules of inference to establish the validity of the argument. "It is not sunny this afternoon and it is colder than yesterday. We will go swimming only if it is sunny. If we do not go swimming, then we will take canoe trip, and if we take a canoe trip, then we will be home by sunset" lead to the conclusion "we will be home by sunset."	6	L3	2	1
b)	Define Euler circuit and Hamiltonian circuit. Give examples.	4	L3	2	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
